

BONSEO KOO

Undergraduate Researcher & M.S. applicant in Electronic Engineering

Department of Electronic Engineering, Sogang University

Seoul, Republic of Korea | kooseo1296@sogang.ac.kr | GitHub: kooseo1296-debug

RESEARCH INTERESTS

- **Hardware Accelerators:** FPGA & Reconfigurable Computing, NPU/AI/LLM Accelerator Architecture
- **Hardware-Efficient AI:** Low-Precision Quantization, Sparse Inference, HW/SW Co-Design
- **Digital Systems:** Digital VLSI Design & Verification, Systolic Array Architectures

EDUCATION

Sogang University — Seoul, Republic of Korea

Bachelor of Science in Electronic Engineering | Expected Graduation: **Aug. 2027**

- **GPA:** 3.49 / 4.30 (Official Percentile: **92.4%**) | Major GPA : **3.79/4.30** | 6 Semesters Completed
- **Selected Advanced Coursework:** Digital Circuit Design (**A+**), Advanced Digital Circuit Design (**A0**), Digital Logic Circuit Laboratory (**A0**), Computer Architecture (**A0**), Probability and Random Variables (**A0**), Semiconductor Device Engineering (Ongoing), VLSI System Design (Ongoing)

PUBLICATIONS & PRESENTATIONS

- **B. Koo**, Y. Lee, T. Jung, and S. Ryu, "Maximizing NPU Fidelity and Efficiency with LOD-Based Dynamic Scaling and Switching-Aware Zero-Skipping," *2026 ISE Summer Conference*, **First-author Oral Presentation**, Jul. 2026.

AWARDS & HONORS

- **Live Demonstration Excellence Award | 2026 ISE Summer Conference** (Jul. 15, 2026)
 - Awarded for the live FPGA demonstration of a low-precision NPU and CIFAR-10 inference pipeline.

RESEARCH & ENGINEERING EXPERIENCE

Digital Systems Design Laboratory, Sogang University — Undergraduate Researcher

Project 1: FPGA NPU Architecture Design & Optimization | Dec. 2025 – Jul. 2026 |
 GitHub: github.com/kooseo1296-debug/Project-1.Energy-Efficient-Low-Precision-NPU

- **Advisor:** Prof. Sungju Ryu
- Designed and implemented an FPGA NPU in Verilog RTL based on an **8×8 weight-stationary systolic array** and matrix-tiling controller; validated at **100 MHz on PYNQ-Z2**.
- Proposed **LOD-based Dynamic Scaling** and **Switching-Aware Zero-Skipping** to optimize the fidelity–area–power trade-off of low-precision inference.
- Evaluated 45 NPU configurations; the selected design achieved **92.0% fidelity** (vs. 74.0% fixed 7-bit baseline) while reducing **LUTs by 23.7%, BRAM by 11.1%, and dynamic power by 12.9%**.
- Integrated the NPU into a **live-camera CIFAR-10 inference pipeline** and presented a live demo at the *2026 ISE Summer Conference*.

Project 2: End-to-End CIFAR-10 FPGA CNN Accelerator | Aug. 2026 - Sep. 2026 |
 GitHub: github.com/kooseo1296-debug/Project-2

- Advisor: Prof. Sungju Ryu
- Re-architected a SmallVGG CIFAR-10 accelerator on PYNQ-Z2, moving Conv1–Conv6, GAP, FC1–FC2 scheduling and intermediate feature processing from the Zynq PS into the PL while retaining input preprocessing on the PS.
- Reduced steady-state PS–PL payload transactions from 748,650 to 3,084 per image (–99.59%) and end-to-end Vitis latency from 338.03 ms to 5.004 ms per image, corresponding to a 67.55× throughput improvement over the PS-managed baseline.
- Preserved CIFAR-10 accuracy at 91.5% versus 91.9% for the baseline using hardware-friendly integer shift-and-round requantization; met timing at 125 MHz with unchanged BRAM and DSP utilization.

- Validated the accelerator on 1,000 CIFAR-10 images and deployed a USB-camera live inference pipeline on PYNQ/Jupyter; separately characterized controlled inference latency and application-level camera/display throughput.

Department of Electrical & Electronic Engineering, Yonsei University — Research Intern

Project: NPU Design and HW/SW Co-Design | Summer 2026

- **Advisor:** Prof. Seungkyu Choi
- Designed a **LeNet-based MNIST inference NPU** on a Zynq SoC, expanding PL-only accelerator design skills to **PS–PL system integration**.
- Developed PS-side control in Vitis; implemented **AXI4-Lite-based memory-mapped control** and **DMA-based high-throughput data transfer** for hardware accelerator integration.
- Studied and presented “*OliVe: Accelerating Large Language Models via Hardware-friendly Outlier-Victim Pair Quantization*”, examining its outlier-victim pair quantization scheme and hardware implications for low-bit LLM inference.

TECHNICAL SKILLS

- **Hardware / RTL:** Verilog HDL, RTL Design & Verification, Systolic Arrays, Fixed-Point Arithmetic, Digital VLSI
- **FPGA / SoC Tools:** AMD/Xilinx Vivado & Vitis, Zynq PS–PL Integration (AXI4-Lite, DMA, BRAM), PYNQ-Z2
- **Programming & ML:** Python, PyTorch, C, Jupyter Notebook
- **Engineering Tools:** MATLAB, Cadence Virtuoso, PSpice, TCAD

LANGUAGES

- **TOEFL iBT: 5.0 / 6.0** (Equivalent to **100/120**, Verified **CEFR C1**)
- *Subscores: Reading 5.0, Listening 5.0, Writing 4.0, Speaking 5.0*